

Modern Algebra

Assignment 6

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6 Group Isomorphisms

Problem 1

Observe that $\phi : \mathbb{Z} \rightarrow 2\mathbb{Z}$ such that $\phi(x) = 2x$ is a bijection:

Suppose $\phi(a) = \phi(b) \implies 2a = 2b \implies a = b$. Thus, the function is clearly injective.

The function is clearly surjective, as for every t in the codomain, by definition, $t = 2k$ where $k \in \mathbb{Z}$. Clearly, $\phi(k) = t$. It is also a homomorphism as

$$\phi(a + b) = 2(a + b) = 2a + 2b = \phi(a) + \phi(b)$$

Problem 2

By Theorem 6.2.2. in the Lecture Notes, if there exist an isomorphism between two cyclic groups, then this isomorphism has to send every generator in the domain to a generator in the codomain. We know that $\mathbb{Z}$ with addition is cyclic and it has only two generators $1, -1$. So we have only two possible isomorphisms from $\mathbb{Z}$ to itself. These two automorphisms are $\phi(x) = x, g(x) = -x$. They are clearly injective and surjective. Observe:

$$\phi(a + b) = a + b = \phi(a) + \phi(b)$$

$$g(a + b) = -(a + b) = -a - b = g(a) + g(b).$$

Both functions are homomorphisms. Hence, ϕ, g are the only possible automorphisms and $\text{Aut}(\mathbb{Z}) = \phi, g$.

Problem 4

By Theorem 6.2.5. in Lecture Notes, if we have an isomorphism between two groups $f : G \rightarrow H$, where one of these groups is cyclic, then the other group must be cyclic. We observe that $\langle 3 \rangle = U_{10}$ but U_8 is not cyclic. Thus, no isomorphism can be found between U_{10}, U_8 .

Problem 8

Proof. We will call the mapping $a \rightarrow \log_{10} a$ by g .

First, we need to show that this function is bijective. We start by showing injectivity. Suppose $g(x) = g(y)$, then $\log_{10} x = \log_{10} y$. We know that the logarithmic function has an inverse, so $x = y$ and g is injective. Next, we show surjectivity. By definition of the codomain, we can see that every element in the codomain can be written as $\log_{10} t = k$, where $t, k \in \mathbb{R}^+, \mathbb{R}$ respectively. Observe that $10^k = t$ and $g(t) = \log_{10} 10^k = k$. Thus, the function is surjective. Therefore, we showed that g is a bijection from $\mathbb{R}^+ \rightarrow \mathbb{R}$.

Second, we will show that g is a homomorphism. Observe

$$g(xy) = \log_{10} xy = \log_{10} x + \log_{10} y \text{ (Using properties of the logarithmic function).}$$

Clearly a homomorphism. Therefore, we showed that the mapping $a \rightarrow \log_{10} a$ is an isomorphism from $\mathbb{R}^+$ under multiplication to $\mathbb{R}$ under addition. ■

Problem 11

$$\begin{aligned} \phi(a^3 b^{-2}) &= \phi(a^3) + \phi(b^{-2}) \\ &= \phi(a \cdot a \cdot a) + \phi(b^{-1} \cdot b^{-1}) \\ &= \phi(a) + \phi(a) + \phi(a) + \phi(b^{-1}) + \phi(b^{-1}) \\ &= 3\bar{a} + \phi(b^{-1}) + \phi(b^{-1}) = 3\bar{a} + 2\phi(b^{-1}). \end{aligned}$$

Since the isomorphism maps inverses to inverses, then $\phi(b^{-1}) = -\phi(b)$. Then the final expression is $3\bar{a} - 2\bar{b}$.

Problem 12

Proof. Assume α is an automorphism, we need to show that G is abelian. Observe since α is a homomorphism, then pick $s, t \in G$:

$$\begin{aligned}\alpha(st) &= (st)^{-1} = t^{-1}s^{-1} \\ \alpha(s)\alpha(t) &= s^{-1}t^{-1}.\end{aligned}$$

And we know that $\alpha(st) = \alpha(s)\alpha(t)$, then $t^{-1}s^{-1} = s^{-1}t^{-1}$. Since, $s, t \in G$ then $s^{-1}, t^{-1} \in G$. then the group is abelian. For the backward direction, assume the group is abelian, we need to show that α is an automorphism. First, we show it is bijective. Since, G is a group, then every element has a unique inverse and α is clearly both injective and surjective. That is, if $\alpha(a) = \alpha(b)$, then $a = b$ as inverses are unique, so α is injective. And since for an element in the codomain a^{-1} , it is also in the group and has a unique inverse a and α is surjective. Second, we show α is a homomorphism, pick $s, t \in G$ and observe:

$$\alpha(st) = (st)^{-1} = t^{-1}s^{-1} = s^{-1}t^{-1} = \alpha(s)\alpha(t).$$

Hence, α is a homomorphism.

Therefore, $\alpha(g) = g^{-1}$ for all $g \in G$ is an isomorphism iff G is abelian. ■

Problem 14

Observe $\mathbb{Z}_1, \mathbb{Z}_2$ are not isomorphic, as $|\mathbb{Z}_1| \neq |\mathbb{Z}_2|$. By theorem 6.3.5. in Lecture Notes, we know that $\text{Aut}(\mathbb{Z}_n) \cong U_n$. So, $\text{Aut}(\mathbb{Z}_2) \cong U_2 = 1$ and $\text{Aut}(\mathbb{Z}_1) \cong U_1 = 1$. Since, $U_1 = U_2$ then $\text{Aut}(\mathbb{Z}_2) \cong U_1$. By transitivity, $\text{Aut}(\mathbb{Z}_1) \cong \text{Aut}(\mathbb{Z}_2)$.

Problem 21

Proof. We see that H contains the permutations that fix 1 and permute the other elements. Since, $\beta \in S_5$ then the remaining elements are $\{2, 3, 4, 5\}$, which are 4 elements. We can see that is isomorphic to S_4 . Similarly, the group K has the permutations that fix 2 and permute $\{1, 3, 4, 5\}$ which is also isomorphic S_4 . Thus, $H \cong S_4 \wedge K \cong S_4 \implies H \cong K$. We use the same approach if $\beta \in S_n$ where $n \geq 3$. We know β fixes an element and permute the others, then the group is isomorphic to S_{n-1} . ■

Problem 22

This is just a generalization to what we did in Problem 1.

Proof. Observe that $\phi : \mathbb{Z} \rightarrow k\mathbb{Z}$ such that $\phi(x) = kx$ is a bijection, where $k \in \mathbb{Z}^*$:

Suppose $\phi(a) = \phi(b) \implies ka = kb \implies a = b$, where $a, b \in \mathbb{Z}$. Thus, the function is clearly injective.

The function is clearly surjective, as for every t in the codomain, by definition, $t = km$ where $m \in \mathbb{Z}$. Clearly, $\phi(m) = t$. It is also a homomorphism as

$$\phi(a + b) = k(a + b) = ka + kb = \phi(a) + \phi(b)$$

Hence, we know that $K\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ for all $k \in \mathbb{Z}^*$ and we showed it is isomorphic to $\mathbb{Z}$. ■

Problem 24

Proof. Since H is the set of the elements fixed by the automorphism ϕ . We can see that the identity element is in H as the automorphism must send the identity to itself. Next, we show it is closed under the operation. Pick $a, b \in H$, we want to show that $ab \in H$. Since ϕ is a homomorphism, then $\phi(ab) = \phi(a)\phi(b)$. But we know that since $a, b \in H$, then ϕ fixes them. Thus, $\phi(ab) = \phi(a)\phi(b) = ab$ and $ab \in H$.

Last, we proof that H is closed under taking inverses. Since ϕ is an isomorphism, then $\phi(a^{-1}) = \phi(a)^{-1} = (\phi(a))^{-1} = a^{-1}$. Hence, it is closed under taking inverses.

Therefore, we showed that H is a subgroup of G . ■

Problem 34

Proof. Since K is a subgroup of G , then it has the identity element. We know that ϕ is an isomorphism from $G \rightarrow \bar{G}$ and it must send the identity element in G to the identity in $\bar{G}$. Thus, we showed $\phi(K)$ is not empty and it has the identity element. Pick elements $a, b \in \phi(K)$ such that $a = \phi(s), b = \phi(t)$, where $s, t \in K$. Observe that $ab = \phi(s)\phi(t) = \phi(st)$ and $st \in K$ since K is a subgroup. Hence, $\phi(K)$ is closed under the operation.

Next, we show that $\phi(K)$ is closed under taking inverses. See that $\phi(s^{-1}) = \phi(s)^{-1} = a^{-1}$, so every element in $\phi(K)$ has an inverse and $\phi(K)$ is closed under taking inverses. Therefore, we proved that $\phi(K)$ is a subgroup of $\bar{G}$ if K is a subgroup of G , where ϕ is an isomorphism from G to $\bar{G}$. ■